

the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1974; No. 539.

- 15 Lewis RJ, ed. *Sax's Dangerous Properties of Industrial Materials*, 11th edn. New York: Wiley, 2004; 309–310.
- 16 Esposito E *et al.* Spray-dried *Eudragit* microparticles as encapsulation devices for vitamin C. *Int J Pharm* 2002; **242**: 329–334.
- 17 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 67.

20 General References

- Abramovici B *et al.* [Comparative study of the tableability of different grades of vitamin C.] *STP Pharma* 1987; **3**: 16–22[in French].
- Allwood MC. Factors influencing the stability of ascorbic acid in total parenteral nutrition infusions. *J Clin Hosp Pharm* 1984; **9**: 75–85.
- Bhagavan HN, Wolkoff BI. Correlation between the disintegration time and the bioavailability of vitamin C tablets. *Pharm Res* 1993; **10**: 239–242.
- Davies MB *et al.* *Vitamin C—Its Chemistry and Biochemistry*. London: Royal Society of Chemistry, 1991.
- Hu F *et al.* Effects of different adhesives on the stability of vitamin C buccal tablets. *Zhejiang Yike Daxue Xuebao* 1997; **26**: 108–110.
- Krishna G *et al.* Development of a parenteral formulation of an investigational anticancer drug, 3-aminopyridine-2-carboxaldehyde thiosemicarbazone. *Pharm Dev Technol* 1999; **4**: 71–80.

Nebuloni M *et al.* Thermal analysis in preformulation studies of a lyophilized form of an antibiotic. *Boll Chim Farm* 1996; **135**: 94–100.

Pinsuwan S *et al.* Degradation kinetics of 4-dedimethylamino sancycline, a new anti-tumor agent, in aqueous solutions. *Int J Pharm* 1999; **181**: 31–40.

Saleh SI, Stamm A. Evaluation of some directly compressible L-ascorbic acid forms. *STP Pharma* 1988; **4**: 10–14.

Saleh SI, Stamm A. Contribution to the preparation of a directly compressible L-ascorbic acid granular form: comparison of granules prepared by three granulation methods and evaluation of their corresponding tablets. *STP Pharma* 1988; **4**: 182–187.

Seta Y *et al.* Preparation and pharmacological evaluation of Captopril sustained-release dosage forms using oily semisolid matrix. *Int J Pharm* 1988; **41**: 255–262.

21 Author

AH Kibbe.

22 Date of Revision

10 February 2009.

Ascorbyl Palmitate

1 Nonproprietary Names

BP: Ascorbyl Palmitate

PhEur: Ascorbyl Palmitate

USP-NF: Ascorbyl Palmitate

2 Synonyms

L-Ascorbic acid 6-palmitate; ascorbylis palmitas; E304; 3-oxo-L-gulofuranolactone 6-palmitate; vitamin C palmitate.

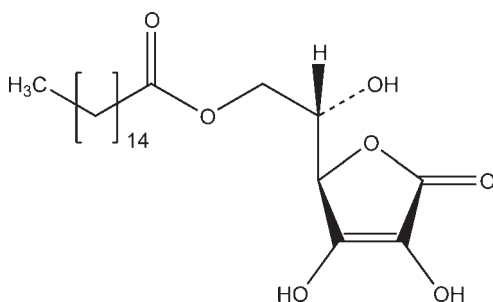
3 Chemical Name and CAS Registry Number

L-Ascorbic acid 6-hexadecanoate [137-66-6]

4 Empirical Formula and Molecular Weight

C₂₂H₃₈O₇ 414.54

5 Structural Formula



6 Functional Category

Antioxidant.

7 Applications in Pharmaceutical Formulation or Technology

Ascorbyl palmitate is primarily used either alone or in combination with alpha tocopherol as a stabilizer for oils in oral pharmaceutical formulations and food products; generally 0.05% w/v is used. It may also be used in oral and topical preparations as an antioxidant for drugs unstable to oxygen. The combination of ascorbyl palmitate with alpha tocopherol shows marked synergism, which increases the effect of the components and allows the amount used to be reduced.

The solubility of ascorbyl palmitate in alcohol permits it to be used in nonaqueous and aqueous systems and emulsions.

8 Description

Ascorbyl palmitate is a practically odorless, white to yellowish powder.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for ascorbyl palmitate.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Appearance of solution	+	—
Melting range	—	107–117°C
Specific rotation (10% w/v in methanol)	+21° to +24°	+21° to +24°
Loss on drying	≤ 1.0%	≤ 2.0%
Residue on ignition	—	≤ 0.1%
Sulfated ash	≤ 0.1%	—
Heavy metals	≤ 10 ppm	≤ 0.001%
Assay (dried basis)	98.0–100.5%	95.0–100.5%

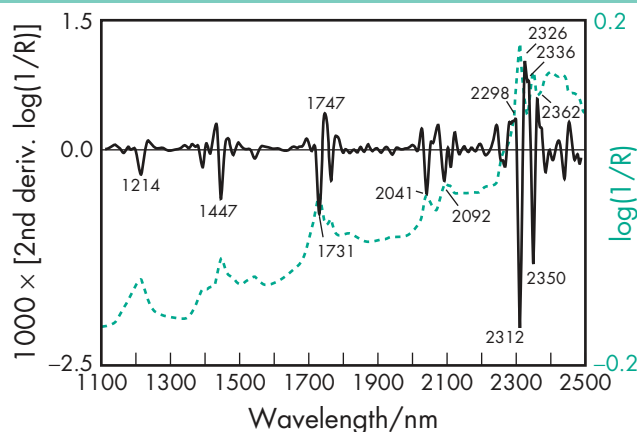


Figure 1: Near-infrared spectrum of ascorbyl palmitate measured by reflectance.

10 Typical Properties

NIR spectra see Figure 1.

Solubility see Table II.

Table II: Solubility of ascorbyl palmitate.

Solvent	Solubility at 20°C unless otherwise stated ⁽¹⁾
Acetone	1 in 15
Chloroform	1 in 3300
Cottonseed oil	1 in 11 at 60°C
Ethanol	1 in 1670
	1 in 8
	1 in 1.7 at 70°C
Ethanol (95%)	1 in 9.3
Ethanol (50%)	1 in 2500
Ether	1 in 132
Methanol	1 in 5.5
	1 in 1.7 at 60°C
Olive oil	1 in 3300
Peanut oil	1 in 3300
Propan-2-ol	1 in 20
	1 in 5 at 70°C
Sunflower oil	1 in 3300
Water	Practically insoluble
	1 in 500 at 70°C
	1 in 100 at 100°C

11 Stability and Storage Conditions

Ascorbyl palmitate is stable in the dry state, but is gradually oxidized and becomes discolored when exposed to light and high humidity. In an unopened container, stored in a cool place, it has a shelf life of at least 12 months. During processing, temperatures greater than 65°C should be avoided.

The bulk material should be stored in an airtight container at 8–15°C, protected from light.

12 Incompatibilities

Incompatibilities are known with oxidizing agents; e.g. in solution oxidation is catalyzed by trace metal ions such as Cu^{2+} and Fe^{3+} .

13 Method of Manufacture

Ascorbyl palmitate is prepared synthetically by the reaction of ascorbic acid with sulfuric acid followed by reesterification with palmitic acid.

14 Safety

Ascorbyl palmitate is used in oral pharmaceutical formulations and food products, and is generally regarded as an essentially nontoxic and nonirritant material. The WHO has set an estimated acceptable daily intake for ascorbyl palmitate at up to 1.25 mg/kg body-weight.⁽²⁾

LD_{50} (mouse, oral): 25 g/kg⁽³⁾

LD_{50} (rat, oral): 10 g/kg

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of material handled. Ascorbyl palmitate dust may cause irritation to the eyes and respiratory tract. Eye protection is recommended.

16 Regulatory Status

GRAS listed. Accepted for use as a food additive in Europe. Included in the FDA Inactive Ingredients Database (oral, rectal, topical preparations). Included in nonparenteral medicines licensed in the UK.

17 Related Substances

Ascorbic acid; sodium ascorbate.

18 Comments

In order to maximize the stability and efficacy of ascorbyl palmitate the following precautions are recommended: stainless steel, enamel, or glass should be used; deaeration (vacuum) procedures and inert gas treatment are recommended where feasible; protect from light and radiant energy.

The formation of ascorbyl palmitate vesicles (Aspasomes) and their pharmaceutical applications has been investigated.⁽⁴⁾

The EINECS number for ascorbyl palmitate is 205-305-4. The PubChem Compound ID (CID) for ascorbyl palmitate is 5282566.

19 Specific References

- 1 Kläui H. Tocopherol, carotene and ascorbyl palmitate. *Int Flavours Food Addit* 1976; 7(4): 165–172.
- 2 FAO/WHO. Toxicological evaluation of certain food additives with a review of general principles and of specifications. Seventeenth report of the joint FAO/WHO expert committee on food additives. *World Health Organ Tech Rep Ser* 1974; No. 539.
- 3 Sweet DV, ed. *Registry of Toxic Effects of Chemical Substances*. Cincinnati: US Department of Health, 1987.
- 4 Gopinath D *et al.* Ascorbyl palmitate vesicles (Aspasomes): formation, characterization and applications. *Int J Pharm* 2004; 271: 95–113.

20 General References

- Austria R *et al.* Stability of vitamin C derivatives in solution and topical formulations. *J Pharm Biomed Anal* 1997; 15: 795–801.
- Daniel JW. Metabolic aspects of antioxidants and preservatives. *Xenobiotica* 1986; 16(10–11): 1073–1078.
- Pongracz G. Antioxidant mixtures for use in food. *Int J Vitam Nutr Res* 1973; 43: 517–525.
- Špiclin P *et al.* Stability of ascorbyl palmitate in topical microemulsions. *Int J Pharm* 2001; 222: 271–279.
- Weller PJ *et al.* Stability of a novel dithranol ointment formulation, containing ascorbyl palmitate as an anti-oxidant. *J Clin Pharm Ther* 1990; 15: 419–423.

21 Author

PJ Weller.

22 Date of Revision

9 January 2009.

1 Nonproprietary Names

BP: Aspartame

PhEur: Aspartame

USP-NF: Aspartame

2 Synonyms

(3S)-3-Amino-4-[[[(1S)-1-benzyl-2-methoxy-2-oxoethyl]amino]-4-oxobutanoic acid; 3-amino-*N*-(α -carboxyphenethyl)succinamic acid *N*-methyl ester; 3-amino-*N*-(α -methoxycarbonylphenethyl)succinamic acid; APM; aspartamum; aspartyl phenylamine methyl ester; *Canderel*; E951; *Equal*; methyl *N*-L- α -aspartyl-L-phenylalaninate; *NutraTaste*; *NutraSweet*; *Pal Sweet*; *Pal Sweet Diet*; *Sanecta*; SC-18862; *Tri-Sweet*.

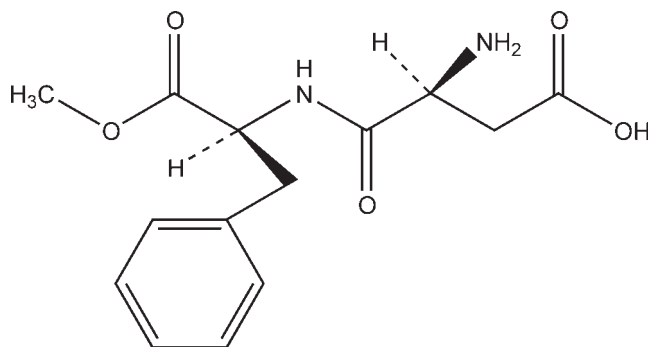
3 Chemical Name and CAS Registry Number

N-L- α -Aspartyl-L-phenylalanine 1-methyl ester [22839-47-0]

4 Empirical Formula and Molecular Weight

C₁₄H₁₈N₂O₅ 294.30

5 Structural Formula



6 Functional Category

Sweetening agent.

7 Applications in Pharmaceutical Formulation or Technology

Aspartame is used as an intense sweetening agent in beverage products, food products, and table-top sweeteners, and in pharmaceutical preparations including tablets,^(1,2) powder mixes, and vitamin preparations. It enhances flavor systems and can be used to mask some unpleasant taste characteristics; the approximate sweetening power is 180–200 times that of sucrose.

Unlike some other intense sweeteners, aspartame is metabolized in the body and consequently has some nutritive value: 1 g provides approximately 17 kJ (4 kcal). However, in practice, the small quantity of aspartame consumed provides a minimal nutritive effect.

8 Description

Aspartame occurs as an off white, almost odorless crystalline powder with an intensely sweet taste.

SEM 1: Excipient: aspartame; magnification: 70 $\times$; voltage: 3 kV.



9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for aspartame.

Test	PhEur 6.0	USP32–NF27
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Conductivity	$\leq 30 \mu\text{S}/\text{cm}$	—
Specific optical rotation	$+14.5^\circ$ to $+16.5^\circ$	$+14.5^\circ$ to $+16.5^\circ$
Related substances	+	—
Heavy metals	$\leq 10 \text{ ppm}$	$\leq 0.001\%$
Loss on drying	$\leq 4.5\%$	$\leq 4.5\%$
Residue on ignition	—	$\leq 0.2\%$
Sulfated ash	$\leq 0.2\%$	—
Impurities	+	—
Transmittance	—	+
Limit of 5-benzyl-3,6-dioxo-2-piperazineacetic acid	—	$\leq 1.5\%$
Chromatographic purity	—	+
Assay	98.0–102.0%	98.0–102.0%

10 Typical Properties

Acidity/alkalinity pH = 4.5–6.0 (0.8% w/v aqueous solution)

Brittle fracture index 1.05⁽³⁾

Bonding index

0.8×10^2 (worst case)⁽³⁾

2.3×10^2 (best case)⁽³⁾

Flowability 44% (Carr compressibility index)⁽³⁾

Density (bulk)

0.5–0.7 g/cm³ for granular grade;

0.2–0.4 g/cm³ for powder grade;

0.17 g/cm³ (Spectrum Quality Products).⁽³⁾

Density (tapped) 0.29 g/cm³ (Spectrum Quality Products)⁽³⁾